

Constrained Latin Hypercube Sampling for Optimal Placement of Rainfall Gauges in the Ecuadorian Amazon

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Background

Rain gauges are the most widely used instrument to quantify precipitation. Precipitation strongly influences the hydrological functioning of the Ecuadorian Amazon, yet its spatio-temporal variability remains poorly understood due to limited number of gauges.

Region of Interest



Figure 1: Ecuador's Amazonia region in green (Ecuador Lives, 2026)

Existing Sites



Figure 2: Research area with the 6 existing sites

Objective

The aim of this study is to analyse optimal rain gauge placement to increase density. Using existing sites, environmental factors, and environmental variability, the appropriate number of additional sampling locations required to represent landscape diversity will be determined, while accounting for accessibility constraints.

Materials & Methods

The following method was modelled after Contreras et al.'s (2019) research on rainfall monitoring network design

- Collect and compress remotely sensed environmental data (PCA)
- Create a cost analysis
- Use conditional Latin Hypercube Sampling (cLHS) to produce optimal rain gauge placement.

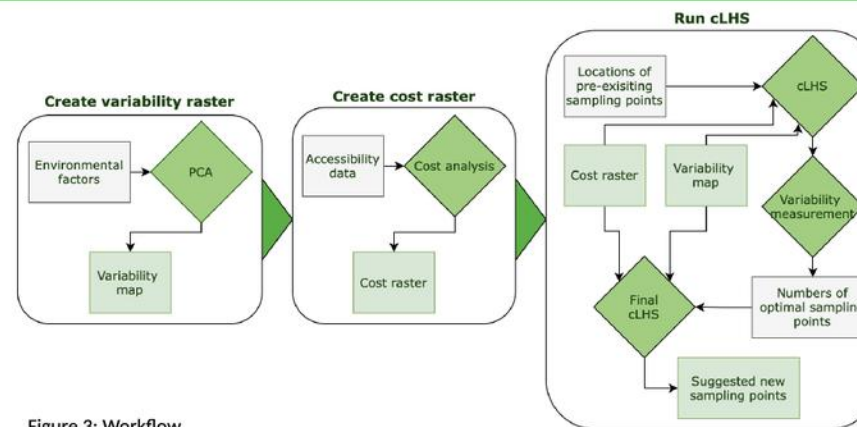


Figure 3: Workflow

Components of the PCA	Components of the Cost raster
Precipitation data	Distance from roads and populated rivers
Monthly average cloud fraction	Slope
NDVI	Protected areas
Total soil Moisture	Intangible zones
World climate data	

Table 1: Components of the PCA and cost raster layers

Optimum sample size was selected by comparing sample standard deviations to the st. d. of values of the entire region (Contreras et al., 2019)

Results

The optimal rain-gauge placement considering accessibility:

- Expanded coverage beyond the western margin
- Sampled previously unrepresented environments
- Captured the full environmental feature space
- Locations with the minimal costs selected

Selected sample size: 23 total stations (6 existing and 17 new)

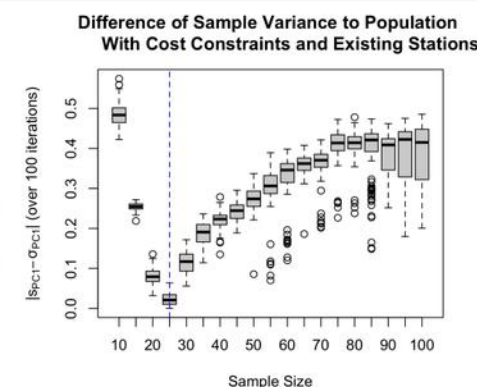


Figure 4: overlay of graphs showing how well samples sizes capture rainfall variation, respectively accounting for cost and preexisting stations and, as in Contreras et al. (2019), without such constraints

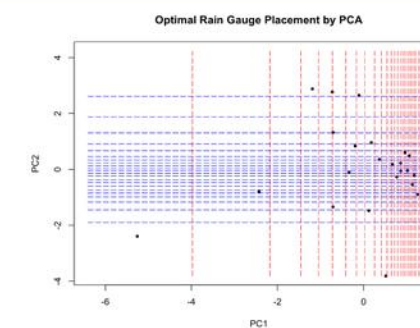


Figure 6: Results of the PCA analysis.

Cost Raster for Accessibility

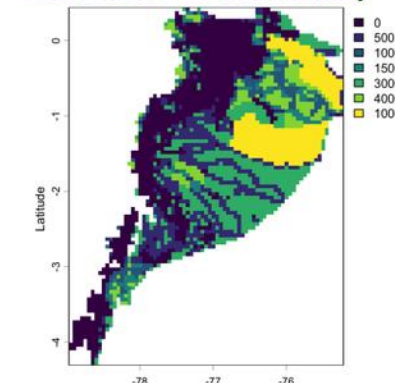


Figure 5: cost raster

Optimized Rain Gauge Network

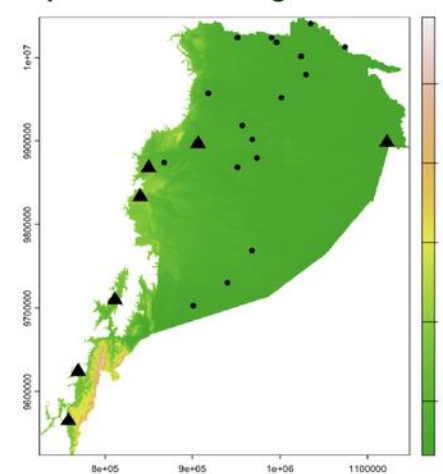


Figure 7: Final locations of existing (▲) and new (●) rain gauges

Discussion & Conclusion

17 new rain gauge locations were selected to most efficiently cover expected spatial rainfall variability in the Ecuadorian Amazon, supplementing the existing stations concentrated in the far west. The sampling is **strongly influenced by cost components incl. national parks & intangible zones** not previously considered by e.g. Contreras et al. (2019)

- There may be more constraints or leeway
- Future research must focus more on costs: optimum sample number is strongly influenced by cost decisions, **higher sample sizes risk overfitting to low-cost areas**

The cLHS is **sensitive to choice of covariates**

- Clear relationship between RS ancillary data and target variable is necessary (e.g. elevation, orographic rainfall, rain shadow, not taken into account here)